

Neuromorphic Shunt Failure Predictor

Executive Technology Brief

Domain: Machine Learning + Clinical Data

Technology Maturity: ENGINEERING_DEFINITION

Transfer Posture: Sponsored Validation

Problem: Shunt failure prediction is reactive (symptoms then surgery)

Mechanism: ML-based predictor using ICP and flow trends to forecast failure ≥ 12 h before symptoms

Architecture: ML predictor using ICP + flow trends to forecast shunt failure 24h before symptoms

Governing Model: Neuromorphic ML predictor for shunt failure prediction. Sensor streams (flow, pressure) processed by feature extractor + ML model to predict imminent failure.

Key Equations:

$P(\text{failure} \mid \text{features}) = \text{ML_model}(\text{features})$ [prediction]

$\text{features} = \text{extract}(\text{raw_sensor_streams})$ [feature extraction]

Alert if $P(\text{failure}) > \text{threshold}$ [decision rule]

Kill Condition: No real failure dataset exists to train the model (CRITICAL BLOCKER per Article XXXVII)

Current State: ENGINEERING_DEFINITION. No physical prototype. No clinical validation.

Disclosure: This dossier distinguishes established evidence from proposals and unresolved questions.